

2025-10-10 - Valentini (4)

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Klimontovich derivation

System of N_0 particles with position $\bar{x}_i(t)$ and velocity $\bar{v}_i(t)$

Equation of motion

$$\begin{cases} \dot{\bar{x}}_i(t) = \bar{v}_i(t) \\ \dot{\bar{v}}_i(t) = \frac{\rho_s}{m_s} \left[\bar{E}_m(\bar{x}_i(t), t) + \frac{\bar{v} \times \bar{B}_m(\bar{x}_i(t), t)}{c} \right] \end{cases}$$

and Maxwell equations

$$\begin{cases} \nabla \cdot \bar{E}_m(\bar{x}_i(t), t) = 4\pi\rho_m(\bar{x}_i(t), t) \\ \nabla \cdot \bar{B}_m(\bar{x}_i(t), t) = 0 \\ \nabla \times \bar{E}_m(\bar{x}_i(t), t) = -\frac{1}{c} \frac{\partial \bar{B}_m(\bar{x}_i(t), t)}{\partial t} \\ \nabla \times \bar{B}_m(\bar{x}_i(t), t) = \frac{4\pi}{c} \bar{J}_m(\bar{x}_i(t), t) + \frac{1}{c} \frac{\partial \bar{E}_m(\bar{x}_i(t), t)}{\partial t} \end{cases}$$

Phase space density N_s

$$N_s(\bar{x}, \bar{v}, t) = \sum_{i=1}^{N_0} \delta(\bar{x} - \bar{x}_i(t)) \delta(\bar{v} - \bar{v}_i(t))$$

Charge density

$$\rho_m(\bar{x}, t) = \sum_s q_s \int N_s(\bar{x}, \bar{v}, t) d\bar{v}$$

Current

$$\bar{J}_m(\bar{x}, t) = \sum_s q_s \int \bar{v} N_s(\bar{x}, \bar{v}, t) d\bar{v}$$

N_s is a "very spiky" function, which enters the Maxwell equations

Let's have a look at some $\delta(x)$ properties:

1. $\int_{-\infty}^{\infty} \delta(x) dx = 1$
2. $\int \delta(x - a) dx = 1, \quad \forall a \in \mathbb{R}$
3. $\int \delta(x - a) f(x) dx = f(a), \quad \forall a \in n\mathbb{R}$
4. $f(a) \delta(a -) = f() \delta(a -), \quad \forall a, \in \mathbb{R}$

Now (using property n.2)

$$\begin{aligned}
N_0 &= \int_{-\infty}^{\infty} N_s(\bar{x}, \bar{v}, t) d\bar{x} d\bar{v} = \\
&= \int_{-\infty}^{\infty} \sum_{i=1}^{N_0} \delta(\bar{x} - \bar{x}_i) \delta(\bar{v} - \bar{v}_i) d\bar{x} d\bar{v} = \\
&= \sum_{i=1}^{N_0} \int_{-\infty}^{\infty} \delta(\bar{x} - \bar{x}_i) d\bar{x} \int_{-\infty}^{\infty} \delta(\bar{v} - \bar{v}_i) d\bar{v}
\end{aligned}$$

We move along the particle trajectories

$$\begin{aligned}
\frac{\partial N_s}{\partial t} &= \sum_{i=1}^{N_0} \frac{\partial N_s}{\partial \bar{x}_i} \frac{\partial \bar{x}_i}{\partial t} + \frac{\partial N_s}{\partial \bar{v}_i} \frac{\partial \bar{v}_i}{\partial t} = \\
&= \sum_{i=1}^{N_0} \left\{ \frac{\partial \bar{x}_i}{\partial t} \frac{\partial}{\partial \bar{x}_i} \left[\sum_{i=1}^{N_0} \delta(\bar{x} - \bar{x}_i) \delta(\bar{v} - \bar{v}_i) \right] + \frac{\partial \bar{v}_i}{\partial t} \frac{\partial}{\partial \bar{v}_i} \left[\sum_{i=1}^{N_0} \delta(\bar{x} - \bar{x}_i) \delta(\bar{v} - \bar{v}_i) \right] \right\}
\end{aligned}$$

Here only the i index survives, while all the mixed-indices terms are zero

$$\frac{\partial N_s}{\partial t} = \sum_{i=1}^{N_0} \dot{\bar{x}}_i \frac{\partial}{\partial \bar{x}_i} \delta(\bar{x} - \bar{x}_i) \delta(\bar{v} - \bar{v}_i) + \dot{\bar{v}}_i \frac{\partial}{\partial \bar{v}_i} \delta(\bar{x} - \bar{x}_i) \delta(\bar{v} - \bar{v}_i)$$

We use the equations of motion

$$\begin{aligned}
\frac{\partial N_s}{\partial t} &= \sum_{i=0}^{N_0} \bar{v}_i \cdot \nabla_{\bar{x}_i} \delta(\bar{x} - \bar{x}_i) \delta(\bar{v} - \bar{v}_i) + \\
&+ \sum_{i=0}^{N_0} \frac{q_s}{m_s} \left[\bar{E}_m(\bar{x}_i, t) + \frac{\bar{v} \times \bar{B}_m(\bar{x}_i, t)}{c} \right] \cdot \nabla_{\bar{v}_i} \delta(\bar{x} - \bar{x}_i) \delta(\bar{v} - \bar{v}_i)
\end{aligned}$$

We exchange the gradient derivative from the particles position to the general coordinate

$$\nabla_{\bar{x}_i} \delta(\bar{x} - \bar{x}_i) = -\nabla_{\bar{x}} \delta(\bar{x} - \bar{x}_i)$$

and the same is valid for $\bar{v}$ and $\bar{v}_i$.

So we have

$$\begin{aligned}
\frac{\partial N_s}{\partial t} &= - \sum_{i=0}^{N_0} \bar{v}_i \cdot \nabla_{\bar{x}} \delta(\bar{x} - \bar{x}_i) \delta(\bar{v} - \bar{v}_i) + \\
&- \sum_{i=0}^{N_0} \frac{q_s}{m_s} \left[\bar{E}_m(\bar{x}_i, t) + \frac{\bar{v} \times \bar{B}_m(\bar{x}_i, t)}{c} \right] \cdot \nabla_{\bar{v}} \delta(\bar{x} - \bar{x}_i) \delta(\bar{v} - \bar{v}_i) = \\
&= - \sum_{i=0}^{N_0} \nabla_{\bar{x}} \cdot [\bar{v}_i \delta(\bar{x} - \bar{x}_i) \delta(\bar{v} - \bar{v}_i)] + \\
&- \sum_{i=0}^{N_0} \nabla_{\bar{v}} \cdot \left\{ \frac{q_s}{m_s} \left[\bar{E}_m(\bar{x}_i, t) + \frac{\bar{v} \times \bar{B}_m(\bar{x}_i, t)}{c} \right] \cdot \delta(\bar{x} - \bar{x}_i) \delta(\bar{v} - \bar{v}_i) \right\}
\end{aligned}$$

Now we use property n.4 two times

$$(\bar{v}_i \times \bar{B}(\bar{x}_i, t) = \bar{v}_i \delta(\bar{v} - \bar{v}_i) \times \bar{B}(\bar{x}_i, t) \delta(\bar{x} - \bar{x}_i))$$

and we get

$$\begin{aligned}
\frac{\partial N_s}{\partial t} &= - \sum_{i=0}^{N_0} \nabla_{\bar{x}} \cdot [\bar{v} \delta(\bar{x} - \bar{x}_i) \delta(\bar{v} - \bar{v}_i)] + \\
&\quad - \sum_{i=0}^{N_0} \nabla_{\bar{v}} \cdot \left\{ \frac{q_s}{m_s} \left[\bar{E}_m(\bar{x}, t) + \frac{\bar{v} \times \bar{B}_m(\bar{x}, t)}{c} \right] \cdot \delta(\bar{x} - \bar{x}_i) \delta(\bar{v} - \bar{v}_i) \right\} = \\
&= -\bar{v} \cdot \nabla_{\bar{x}} \left[\sum_{i=0}^{N_0} \delta(\bar{x} - \bar{x}_i) \delta(\bar{v} - \bar{v}_i) \right] + \\
&\quad - \frac{q_s}{m_s} \left[\bar{E}_m(\bar{x}, t) + \frac{\bar{v} \times \bar{B}_m(\bar{x}, t)}{c} \right] \cdot \nabla_{\bar{v}} \left[\sum_{i=0}^{N_0} \delta(\bar{x} - \bar{x}_i) \delta(\bar{v} - \bar{v}_i) \right]
\end{aligned}$$

Using the definition of N_s we get the **Klimontovich equation**

$$\frac{\partial N_s}{\partial t} + \bar{v} \cdot \nabla_{\bar{x}} N_s + \frac{q_s}{m_s} \left[\bar{E}_m(\bar{x}, t) + \frac{\bar{v} \times \bar{B}_m(\bar{x}, t)}{c} \right] \cdot \nabla_{\bar{v}} N_s = 0$$

This equation is

- 7 dimensional
- non-linear
- EXACT (and impossible to solve)
- with $\delta(x)$ functions

Note:

$$\frac{dN_s}{dt}_{\text{ort}} = \frac{\partial N_s}{\partial t} + \frac{\partial N_s}{\partial \bar{x}} \frac{\partial \bar{x}}{\partial t} + \frac{\partial N_s}{\partial \bar{v}} \frac{\partial \bar{v}}{\partial t} = 0$$

with the partial derivatives being the terms of Klimontovich equation.

We now use the ensemble average

$$f_s(\bar{x}, \bar{v}, t) \quad N_s(\bar{x}, \bar{v}, t)$$

so we can write

$$N_s(\bar{x}, \bar{v}, t) = f_s(\bar{x}, \bar{v}, t) + \delta N_s, \quad \delta N_s = 0$$

same goes for $\bar{E}$ and $\bar{B}$

$$\bar{E}_m = \bar{E} + \delta \bar{E}, \quad \delta \bar{E} = 0$$

$$\bar{B}_m = \bar{B} + \delta \bar{B}, \quad \delta \bar{B} = 0$$

Now, substituting into Maxwell equations

We start from

$$\nabla \cdot \bar{E}_m = 4\pi\rho_m$$

$$\nabla \cdot (\bar{E} + \delta \bar{E}) = \nabla \cdot \bar{E}$$

because derivatives ($\nabla \cdot$) do not depend on the ensemble ()

$$\begin{aligned}
 \rho_m(\bar{x}, t) &= \sum_s q_s \int N_s(\bar{x}, \bar{v}, t) d\bar{v} = \\
 &= \sum_s q_s \int (f_s(\bar{x}, \bar{v}, t) + \delta N_s) d\bar{v} = \\
 &= \sum_s q_s \int \left(f_s(\bar{x}, \bar{v}, t) + \cancel{\delta N_s} \right) d\bar{v} = \\
 &= \sum_s q_s \int f_s(\bar{x}, \bar{v}, t) d\bar{v}
 \end{aligned}$$

so

$$\nabla \cdot \bar{E} = 4\pi \sum_s q_s \int f_s d\bar{v}$$

Little recap

1. we take the maxwell equations
2. we substitute the ensemble averaged quantities and the "perturbation"
3. we evaluate the ensemble average
4. we eliminate the ensemble averaged "perturbation"

for the other Maxwell equation we obtain

$$\left\{ \begin{aligned}
 \nabla \cdot \bar{B} &= 0 \\
 \nabla \times \bar{E} &= -\frac{1}{c} \frac{\partial \bar{B}}{\partial t} \\
 \nabla \times \bar{B} &= \frac{4\pi}{c} \bar{J} + \frac{\partial \bar{E}}{\partial t} = \\
 &= \frac{4\pi}{c} \sum_s q_s \int f_s v d\bar{v} + \frac{1}{c} \frac{\partial \bar{E}}{\partial t}
 \end{aligned} \right.$$

From the Klimontovich equation we have

$$\frac{\partial}{\partial t}(f_s + \delta N_s) + \bar{v} \cdot \nabla (f_s + \delta N_s) + \frac{q_s}{m_s} \left(\bar{E} + \delta \bar{E} \right) + \frac{\bar{v} \times (\bar{B} + \delta \bar{B})}{c} \cdot \nabla_{\bar{v}} (f_s + \delta N_s) = 0$$

and we take the ensemble average

$$\begin{aligned}
0 &= \frac{\partial f_s}{\partial t} + \nabla f_s + \frac{q_s}{m_s} \left[\bar{E} \cdot \nabla_{\bar{v}} f_s + \cancel{\bar{E} \cdot \nabla_{\bar{v}} \delta N_s} + \cancel{\delta \bar{E} \cdot \nabla_{\bar{v}} f_s} + \delta \bar{E} \cdot \nabla_{\bar{v}} \delta N_s + \right. \\
&\quad \left. + \frac{\bar{v} \times \bar{B}}{c} \nabla_{\bar{v}} f_s + \cancel{\frac{\bar{v} \times \bar{B}}{c} \nabla_{\bar{v}} \delta N_s} + \cancel{\frac{\bar{v} \times \delta \bar{B}}{c} \nabla_{\bar{v}} f_s} + \frac{\bar{v} \times \delta \bar{B}}{c} \nabla_{\bar{v}} \delta N_s \right] = \\
&= \frac{\partial f_s}{\partial t} + \nabla f_s + \frac{q_s}{m_s} \left[\bar{E} \cdot \nabla_{\bar{v}} f_s + \delta \bar{E} \cdot \nabla_{\bar{v}} \delta N_s + \frac{\bar{v} \times \bar{B}}{c} \nabla_{\bar{v}} f_s + \frac{\bar{v} \times \delta \bar{B}}{c} \nabla_{\bar{v}} \delta N_s \right] = \\
&= \frac{\partial f_s}{\partial t} + \nabla f_s + \frac{q_s}{m_s} \left[\bar{E} \cdot \nabla_{\bar{v}} f_s + \frac{\bar{v} \times \bar{B}}{c} \nabla_{\bar{v}} f_s + \delta \bar{E} \cdot \nabla_{\bar{v}} \delta N_s + \frac{\bar{v} \times \delta \bar{B}}{c} \nabla_{\bar{v}} \delta N_s \right]
\end{aligned}$$

The following two terms

$$\delta \bar{E} \cdot \nabla_{\bar{v}} \delta N_s \quad \text{an} \quad \frac{\bar{v} \times \delta \bar{B}}{c} \nabla_{\bar{v}} \delta N_s$$

are two non-linear terms.

To answer why they are not zero, let's assume that every perturbation is a sine wave, we will have

$$\sin = 0 \quad \text{t} \quad \sin \sin \phi \neq 0$$

The equation is usually rewritten as

$$\frac{\partial f_s}{\partial t} + \nabla f_s + \frac{q_s}{m_s} \left[\bar{E} + \frac{\bar{v} \times \bar{B}}{c} \right] \cdot \nabla_{\bar{v}} f_s = -\frac{q_s}{m_s} \left[\delta \bar{E} + \frac{\bar{v} \times \delta \bar{B}}{c} \right] \cdot \nabla_{\bar{v}} \delta N_s$$

or

$$\frac{\partial f_s}{\partial t} + \nabla f_s + \frac{q_s}{m_s} \left[\bar{E} + \frac{\bar{v} \times \bar{B}}{c} \right] \cdot \nabla_{\bar{v}} f_s = \frac{\partial f}{\partial t}_{\text{cosons}} \approx 0$$

the right term contains the granularity of the system and it is often neglected.

In plasma physics we have very little collisions, so we can safely assume that this term is ≈ 0 .

We finally got the Vlasov Maxwell equations.

Linearized Vlasov-Poisson system

So, we used

- electrostatic approximation ($\bar{E} = -\nabla \phi$)
- unmagnetized plasma ($\bar{B} = 0$)

so the polished Vlasov Maxwell equations becomes the set of **Vlasov Poisson equations**

$$\begin{cases} \frac{\partial f_\alpha}{\partial t} + \bar{v} \cdot \nabla f_\alpha + \frac{q_\alpha}{m_\alpha} \left[\underbrace{\bar{E}}_{-\nabla \phi} + \cancel{\frac{\bar{v} \times \bar{B}}{c}} \right] \cdot \nabla_{\bar{v}} f_\alpha = 0 \\ -\nabla^2 \phi = 4\pi \sum_\alpha q_\alpha \int_{-\infty}^{\infty} f_\alpha(\bar{r}, \bar{v}, t) d^3v \end{cases}$$

At equilibrium we have Maxwell distribution

$$f_{\alpha 0} = \frac{n_{\alpha 0} m_\alpha^{3/2}}{(2\pi k_B T_{\alpha 0})^{3/2}} \exp \left[-\frac{m_\alpha v^2}{2k_B T_{\alpha 0}} \right]$$

and

$$\begin{aligned} n_{\alpha 0} &= \text{const}, & \bar{U}_{\alpha 0} &= 0, & T_{\alpha 0} &= \text{const} \\ E_0 &= 0, & n_{i0} = n_{e0} &= n_0, & \phi_0 = K &= 0 \end{aligned}$$

without any loss of generality.

Now we perturb the system

$$\begin{cases} f_{\alpha}(\bar{r}, \bar{v}, t) = f_{\alpha 0}(\bar{v}) + f_{\alpha 1}(\bar{r}, \bar{v}, t) \\ \phi(\bar{r}, t) = \phi_1(\bar{r}, t) \end{cases}$$

We substitute in the Vlasov equation and we have

$$\frac{\partial}{\partial t} (\cancel{f_{\alpha 0}} + f_{\alpha 1}) + \bar{v} \cdot \nabla (\cancel{f_{\alpha 0}} + f_{\alpha 1}) - \frac{q_{\alpha}}{m_{\alpha}} \nabla \phi_1 \cdot \nabla_{\bar{v}} (f_{\alpha 0} + f_{\alpha 1}) = 0$$

If we neglected the nonlinear term

$$- \frac{q_{\alpha}}{m_{\alpha}} \nabla \phi_1 \cdot \nabla_{\bar{v}} f_{\alpha 1}$$

we get the linearized form of the Vlasov equation.

$$\frac{\partial}{\partial t} f_{\alpha 1} + \bar{v} \cdot \nabla f_{\alpha 1} - \frac{q_{\alpha}}{m_{\alpha}} \nabla \phi_1 \cdot \nabla_{\bar{v}} f_{\alpha 0} = 0$$

Now we substitute in Poisson equation

$$\begin{aligned} -\nabla^2 \phi &= 4\pi \sum_{\alpha} q_{\alpha} \int_{-\infty}^{\infty} f_{\alpha}(\bar{r}, \bar{v}, t) d^3 v = \\ &= 4\pi \sum_{\alpha} q_{\alpha} \int_{-\infty}^{\infty} (f_{\alpha 0} + f_{\alpha 1}) d^3 v = \\ &= 4\pi e \underbrace{\int_{-\infty}^{\infty} f_{i0} d^3 v}_{n_{i0}} - e \underbrace{\int_{-\infty}^{\infty} f_{e0} d^3 v}_{n_{e0}} + 4\pi \sum_{\alpha} q_{\alpha} \int_{-\infty}^{\infty} f_{\alpha 1} d^3 v = \\ &= 4\pi \sum_{\alpha} q_{\alpha} \int_{-\infty}^{\infty} f_{\alpha 1} d^3 v \end{aligned}$$

where we used the fact that $n_{i0} = n_{e0}$ to eliminate the term with $f_{\alpha 0}$.

So the linearized form for the Vlasov Poisson system is

$$\begin{cases} \frac{\partial}{\partial t} f_{\alpha 1} + \bar{v} \cdot \nabla f_{\alpha 1} - \frac{q_{\alpha}}{m_{\alpha}} \nabla \phi_1 \cdot \nabla_{\bar{v}} f_{\alpha 0} = 0 \\ -\nabla^2 \phi = 4\pi \sum_{\alpha} q_{\alpha} \int_{-\infty}^{\infty} f_{\alpha 1} d^3 v \end{cases}$$

Vlasov solution (not the correct approach to solve this)

Vlasov solved this system using the Fourier transform

$$\begin{cases} f_{\alpha 1}(\bar{r}, \bar{v}, t) = \hat{f}_{\alpha 1}(\bar{v}) \exp \left[i \left(\bar{k} \cdot \bar{r} - \omega t \right) \right] \\ \phi_1(\bar{r}, t) = \hat{\phi}_1 \exp \left[i \left(\bar{k} \cdot \bar{r} - \omega t \right) \right] \end{cases}$$

and

$$\begin{cases} -i\omega \hat{f}_{\alpha 1}(\bar{v}) + \bar{v} \cdot i\bar{k} \hat{f}_{\alpha 1}(\bar{v}) - \frac{q_\alpha}{m_\alpha} i\bar{k} \hat{\phi}_1 \cdot \nabla_{\bar{v}} f_{\alpha 0} = 0 \\ k^2 \hat{\phi}^2 = 4\pi \sum_{\alpha} q_\alpha \int_{-\infty}^{\infty} \hat{f}_{\alpha 1}(\bar{v}) d^3v \end{cases}$$

From Vlasov equation we have

$$\hat{f}_{\alpha 1} = \frac{-\frac{q_\alpha}{m_\alpha} \hat{\phi}_1 \bar{k} \cdot \nabla_{\bar{v}} f_{\alpha 0}}{\omega - \bar{k} \cdot \bar{v}}$$

and using it in Poisson equation

$$\begin{aligned} k^2 \hat{\phi}_1 &= 4\pi \sum_{\alpha} q_\alpha \int_{-\infty}^{\infty} \frac{-\frac{q_\alpha}{m_\alpha} \hat{\phi}_1 \bar{k} \cdot \nabla_{\bar{v}} f_{\alpha 0}}{\omega - \bar{k} \cdot \bar{v}} d^3v = \\ &= -4\pi \sum_{\alpha} \frac{k^2 n_{\alpha 0}}{k^2 n_{\alpha 0}} \cdot \frac{q_\alpha^2}{m_\alpha} \hat{\phi}_1 \int_{-\infty}^{\infty} \left[\frac{\bar{k} \cdot \nabla_{\bar{v}} f_{\alpha 0}}{\omega - \bar{k} \cdot \bar{v}} \right] d^3v = \\ &= -k^2 \hat{\phi}_1 \sum_{\alpha} \frac{1}{k^2 n_{\alpha 0}} \cdot \frac{4\pi q_\alpha^2 n_{\alpha 0}}{m_\alpha} \int_{-\infty}^{\infty} \left[\frac{\bar{k} \cdot \nabla_{\bar{v}} f_{\alpha 0}}{\omega - \bar{k} \cdot \bar{v}} \right] d^3v = \\ &= -k^2 \hat{\phi}_1 \sum_{\alpha} \frac{\omega_{p\alpha}^2}{k^2 n_{\alpha 0}} \int_{-\infty}^{\infty} \left[\frac{\bar{k} \cdot \nabla_{\bar{v}} f_{\alpha 0}}{\omega - \bar{k} \cdot \bar{v}} \right] d^3v \end{aligned}$$

and finally

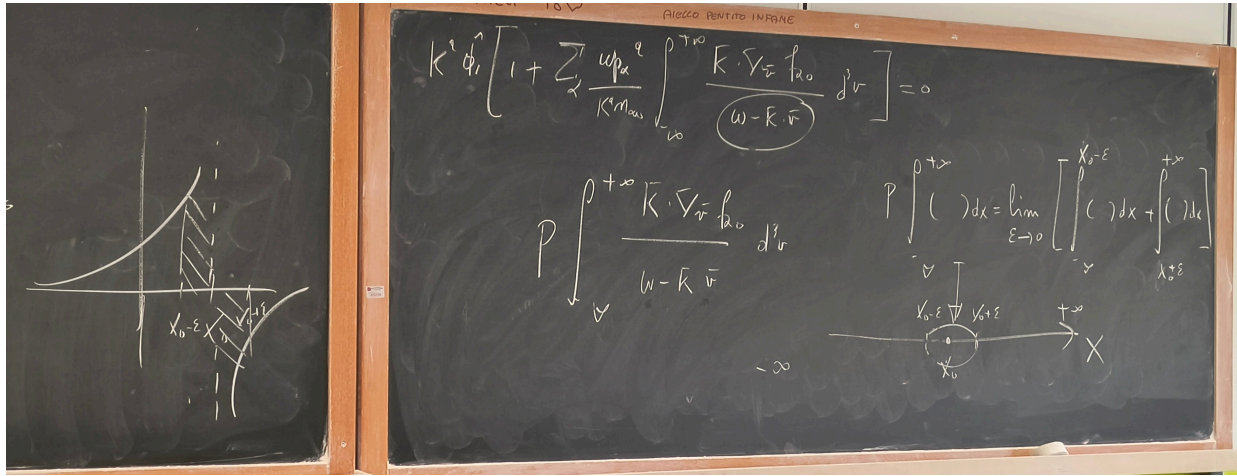
$$k^2 \hat{\phi}_1 \left[1 + \sum_{\alpha} \frac{\omega_{p\alpha}^2}{k^2 n_{\alpha 0}} \int_{-\infty}^{\infty} \left[\frac{\bar{k} \cdot \nabla_{\bar{v}} f_{\alpha 0}}{\omega - \bar{k} \cdot \bar{v}} \right] d^3v \right] = 0$$

Now, the cases for $k^2 = 0$ or $\hat{\phi}_1 = 0$ are not interesting

The integral in this expression is an integral with a divergence in $\omega = \bar{k} \cdot \bar{v}$

Vlasov took the principal part of the integral missing all the important information which happen close to the pole of the integral.

pic 01 - principal part



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